

What is KMP?

Kotlin MultiPlatform(KMP) is a feature released in 2017 with Kotlin 1.2. This feature allows developers to write shared code that can be used across multiple platforms. It enables code reuse between different platforms, such as Android, iOS, web, desktop, and others. With KMP, you can write common business logic, data models, algorithms, and other core components in Kotlin and share them across platforms. This reduces the need for writing separate codebases for each platform, saving development time and effort.

Features of KMM:-

1. Shared code:- KMM allows developers to share business logic, data models, networking code, and even UI-related code between Android and iOS platforms using Kotlin.
2. Platform-specific code:- KMM provides the flexibility to write platform-specific code, which allows developers to leverage platform-specific features, APIs, and UI components of Android and iOS.
3. Tooling and IDE support:- KMM tools include productivity features for developers like integration with Android Studio and Xcode, assistance with debugging, code analysis, and more.
4. Unified build system:- KMM unified build system allows developers to build, test, and deploy shared code and platform-specific code using a single set of build tools.

Architecture of KMP

Using Kotlin Multiplatform, you can share common code across multiple platforms including Android, iOS, web, desktop, and others. Multiplatform libraries provide the necessary APIs for common Kotlin code and allow the writing of shared parts of projects in Kotlin all in one place.

Kotlin compiler has 2 components : Frontend and Backend.

Frontend Compiler:-

The frontend compiler in Kotlin Multiplatform is responsible for processing the Kotlin source code and generating an intermediate representation called the Kotlin Intermediate Representation (KIR).

The frontend compiler performs syntax and semantic analysis, type checking, and other language-specific operations.

Backend Compiler:-

The backend compilers in Kotlin Multiplatform are responsible for taking the KIR generated by the frontend compiler and producing platform-specific code that can run on the target platforms.

There are three different backend compilers: one for each of the Java virtual machine (JVM), JavaScript (JS) and native targets.

What is cloud computing?

Cloud computing means that all the computing hardware and software resources that you need to process your tasks are provided for you, "as a service" over the internet, by a vendor instead of you owning and maintaining them.

It is the duty of the vendor, the Cloud Service Provider (CSP), to develop, own and maintain these resources and make them accessible to the consumers over the internet.

What is rapid elasticity

Cloud services provide the consumer, with the option of quickly "stretching" and "releasing" the capacity of the computing resources as per their needs at any point of time.

In the following pages let us see how other cloud capabilities, namely "on-demand self-service" and "resource pooling", enable rapid elasticity.

What is on-demand self service

As a cloud consumer, you need not depend on the CSP whenever you want to add or release resources. Most CSPs provide a self-service portal, which is interfaced with automated resource management systems. This enables you to instantly (on-demand) add, release, track and pay for your resources, without the intervention of the CSPs.

What is resource pooling

Cloud service providers maintain a large pool of hardware and software resources. For efficient utilization of these computing resources, cloud service providers allocate the same physical resources across multiple cloud consumers, by securely separating it on a logical level. The resources are allocated either automatically, based on their demand, using load balancers, or manually, based on their requests, through self-provisioning portals.

What is metered billing

A cloud service provider measures and meters the usage of resources by a consumer. This is accomplished by including built-in services that tracks the quantity of various IT resources a consumer uses.

Usually, you need not pay upfront or commit to any minimum resource allocation. You must pay at the end of the usage period, only for the number of units of resources that were used. This model of cloud service billing is the most popular and it's called Pay As You Go (PAYG).